

Dimensional Analysis and Significant Figures

Rajat Kalia

15 August, 2026

Infrastructure

#764, 41-A, Chandigarh {TC(official)} - 160036
 Shamlat (Ramgarh Sardaran), Ludhiana, Punjab
 SCO 24(TF), 41-D, Chandigarh - 160036
<https://www.alphaclasses.com/>

Educational

<https://github.com/rajatkalia1982>
<https://sourceforge.net/u/rajatkalia/profile>
<https://www.youtube.com/@rajatkaliaeducation>

Store

<https://www.lulu.com/spotlight/rajatkalia11>
<https://amazon.com/author/rajatkalia>

Sports

<https://www.youtube.com/@indiangamersorganization>
<https://www.facebook.com/groups/igohn>
<https://india.voobly.com/>

Social

<https://www.goodreads.com/sardarbhagatsingh>
<https://x.com/RajatKalia1>
<https://www.linkedin.com/in/rajatkalia007/>
<https://www.facebook.com/rajatkalia007>
<https://www.instagram.com/rajatkalia007/>
<https://www.twitch.tv/rajatkalia>
<https://www.facebook.com/rajatkaliaeducation>
<https://www.researchgate.net/profile/Rajat-Kalia>
<https://www.w3profile.com/rajatkalia/>
<https://www.reddit.com/user/rajatkalia/>

Blogs

<https://rk-e.blogspot.in/> [Physics]
<https://chem-rke.blogspot.com/> [Chemistry]
<https://biosciences-rke.blogspot.com/> [Bio-Sciences]
<https://rk-edu.blogspot.com/> [Mathematics]
<https://medlife-rke.blogspot.com/> [Medical Sciences]
<https://computers-mke.blogspot.com/> [Computers Engineering]
<https://ee-mke.blogspot.com/> [Electrical Engineering]
<https://me-mke.blogspot.com/> [Mechanical Engineering]
<https://ce-mke.blogspot.com/> [Civil Engineering]
<https://mke-whj.blogspot.com/> [White Hat Junior]
<https://ba-xpn.blogspot.com/> [Bachelor of Arts]
<https://constitution1982.blogspot.com/> [Constitution]
<https://debk-mke.blogspot.com/> [Accounting]
<https://comm-pke.blogspot.com/> [Commerce]
<https://brigade-13.blogspot.com/> [Air Force, India]
<https://battalion-7.blogspot.com/> [Armed Forces]
<https://design-mke.blogspot.com/> [Design]
<https://tmm18.blogspot.com/> [Meditation]

Freelancing

<https://www.guru.com/freelancers/rajat-kalia>
<https://www.codecademy.com/profiles/rajatkalia>
<https://leetcode.com/u/rajatkalia1/>
<https://www.scribd.com/user/308680256/Rajat-Kalia>

Contents

I	Dimensional Analysis	3
1	JEE Mains Problems	5
1.1	Intext Problems	5
1.2	Full Syllabus Problems	5
2	NEET	6
II	Significant Figures	7
3	JEE Mains Problems	9

Part I

Dimensional Analysis

Problem Set

Chapter 1

JEE Mains Problems

1.1 Intext Problems

Q1: The dimensional formula of $\frac{1}{2}\epsilon_0 E^2$ (ϵ_0 = permittivity of vacuum and E = Electric Field) is $M^a L^b T^c$.
The value of $2a - b + c = 0$ is _____.

[JEE Mains 2nd April 2026 Shift 1 Q26]

- A) 0
- B) 1
- C) -1
- D) 2

{Solution : <https://www.youtube.com/watch?v=CAgTYRH7-P8> }

1.2 Full Syllabus Problems

Q1: The position of an object having mass 0.1 kg as a function of time t is given as $\vec{r} = 10t^2\hat{i} + 5t^3\hat{j}$ m. At $t = 1$ s, which of the following statements is correct ?

[JEE Mains 2nd April 2026 Shift 1 Q29]

A The linear momentum $\vec{p} = (2\hat{i} + 1.5\hat{j})kg.m/s$.

B The force acting on the object $\vec{F} = (2\hat{i} + 3\hat{j}) N$.

C The angular momentum of the object about it's origin $\vec{L} = 15\hat{k} J.s$.

D The torque acting on the object about it's origin $T = 20\hat{k} N.m$.

Choose the correct answer from the options given below :

- (a) A, B and C only.
- (b) B, C and D only.
- (c) A, C and D only.
- (d) A, B and D only.

{Solution : <https://www.youtube.com/watch?v=DvIERvObc1E>}

Chapter 2

NEET

Q1: Dimensions of stress are :

[NEET E1 Q167]

(1) MLT^{-2}

(2) ML^2T^{-2}

(3) ML^0T^{-2}

(4) $ML^{-1}T^{-2}$

{Solution : <https://www.youtube.com/watch?v=5ewh2C2yY>}

Part II

Significant Figures

Problem Set

Chapter 3

JEE Mains Problems

Q1 : The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm. The length measured by a scale of least count 0.1 cm is 150 cm. When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm, measured by a micrometer of least count 0.001 cm. The error in the measurement of Young's Modulus is $\alpha \times 10^9 \text{ N/m}^2$. The value of α is _____.

(Ignore the contribution of load to Young's Modulus error calculation)

[JEE Mains 2nd April 2026 Shift 1 Q27]

- A) 1.3
- B) 1.65
- C) 0.13
- D) 0.25

{Solution : <https://www.youtube.com/watch?v=JCaMvvfacDw>}